

Event Study

Event Study

```
library(readr)
library(dplyr)
library(fixest)

did_panel_candidates <- c(
  "../..../results/data/did_panel_monthly.csv",
  "../..../did_panel_monthly.csv"
)
did_panel_path <- did_panel_candidates[file.exists(did_panel_candidates)][1]
if (is.na(did_panel_path)) {
  stop("Could not find did_panel_monthly.csv in expected paths.")
}

figures_dir <- "../..../results/figures/event_study"
dir.create(figures_dir, recursive = TRUE, showWarnings = FALSE)

seasonality_dir <- "../..../results/figures/seasonality"
dir.create(seasonality_dir, recursive = TRUE, showWarnings = FALSE)

tables_dir <- "../..../results/tables/event_study"
dir.create(tables_dir, recursive = TRUE, showWarnings = FALSE)

df <- read_csv(did_panel_path, show_col_types = FALSE) %>%
  mutate(
    product = as.character(hs8),
    date = as.Date(date)
  ) %>%
  filter(
    !is.na(product), !is.na(date),
```

```

    !is.na(post), !is.na(highgap)
  )

df_customs <- df %>% filter(!is.na(ln_customs_value))
df_qty <- df %>% filter(!is.na(ln_first_unit_qty))
df_uv <- df %>% filter(!is.na(ln_unit_value))

cat("Using panel file:", did_panel_path, "\n")

```

Using panel file: ../../results/data/did_panel_monthly.csv

```

WIN_PRE <- 60
WIN_POST <- 48

plot_event_study_year <- function(model_obj, output_png, y_label, win_pre, win_post, term_name) {
  b <- coef(model_obj)
  s <- se(model_obj)

  coef_pattern <- paste0("^event_time::[-]?[0-9]+:", term_name, "$")
  parse_pattern <- paste0("^event_time::(-?[0-9]+):", term_name, "$")

  keep <- grepl(coef_pattern, names(b))
  if (!any(keep)) {
    stop("No event-study coefficients matched expected naming pattern.")
  }

  event_time <- as.integer(sub(parse_pattern, "\\1", names(b)[keep]))
  est <- as.numeric(b[keep])
  stderr <- as.numeric(s[keep])
  ci_low <- est - 1.96 * stderr
  ci_high <- est + 1.96 * stderr

  ord <- order(event_time)
  event_time <- event_time[ord]
  est <- est[ord]
  ci_low <- ci_low[ord]
  ci_high <- ci_high[ord]

  tick_min <- floor(min(event_time) / 12) * 12
  tick_max <- ceiling(max(event_time) / 12) * 12
  tick_at <- seq(tick_min, tick_max, by = 12)

```

```

tick_labels <- base_year + tick_at / 12

png(output_png, width = 1000, height = 700)
plot(
  event_time, est,
  pch = 16,
  col = "black",
  xlab = "Year",
  ylab = y_label,
  xaxt = "n",
  ylim = range(c(ci_low, ci_high, 0), na.rm = TRUE)
)
segments(event_time, ci_low, event_time, ci_high, col = "gray40")
lines(event_time, est, col = "black")
axis(1, at = tick_at, labels = tick_labels)
abline(h = 0, lty = 2, col = "red")
dev.off()
}

plot_event_study_year_guides <- function(model_obj, output_png, y_label, win_pre, win_post, t) {
  b <- coef(model_obj)
  s <- se(model_obj)

  coef_pattern <- paste0("^event_time::[-]?[0-9]+:", term_name, "$")
  parse_pattern <- paste0("^event_time::(-?[0-9]+):", term_name, "$")

  keep <- grepl(coef_pattern, names(b))
  if (!any(keep)) {
    stop("No event-study coefficients matched expected naming pattern.")
  }

  event_time <- as.integer(sub(parse_pattern, "\\1", names(b)[keep]))
  est <- as.numeric(b[keep])
  stderr <- as.numeric(s[keep])
  ci_low <- est - 1.96 * stderr
  ci_high <- est + 1.96 * stderr

  ord <- order(event_time)
  event_time <- event_time[ord]
  est <- est[ord]
  ci_low <- ci_low[ord]
  ci_high <- ci_high[ord]

```

```

tick_min <- floor(min(event_time) / 12) * 12
tick_max <- ceiling(max(event_time) / 12) * 12
tick_at <- seq(tick_min, tick_max, by = 12)
tick_labels <- base_year + tick_at / 12

png(output_png, width = 1000, height = 700)
plot(
  event_time, est,
  type = "n",
  xlab = "Year",
  ylab = y_label,
  xaxt = "n",
  ylim = range(c(ci_low, ci_high, 0), na.rm = TRUE)
)

usr <- par("usr")
y_bottom <- usr[3]
y_top <- usr[4]

if (length(tick_at) >= 2) {
  for (i in seq_len(length(tick_at) - 1)) {
    if (i %% 2 == 0) {
      rect(tick_at[i], y_bottom, tick_at[i + 1], y_top, col = "gray95", border = NA)
    }
  }
}
abline(v = tick_at, col = "gray85", lwd = 0.8)

segments(event_time, ci_low, event_time, ci_high, col = "gray40")
points(event_time, est, pch = 16, col = "black")
lines(event_time, est, col = "black")
axis(1, at = tick_at, labels = tick_labels)
abline(h = 0, lty = 2, col = "red")
mtext("Shaded bands and vertical lines mark 12-month intervals", side = 3, line = 0.2, cex = 0.8)
dev.off()
}

es_customs <- df_customs %>%
  filter(!is.na(event_time)) %>%
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)

cat("Rows in customs event-study sample:", nrow(es_customs), "\n")

```

Rows in customs event-study sample: 746784

```
cat("Customs event-time range:", min(es_customs$event_time), "to", max(es_customs$event_time))
```

Customs event-time range: -60 to 47

```
m_es_customs_base <- feols(
  ln_customs_value ~ i(event_time, highgap, ref = -1) | product + date,
  data = es_customs,
  vcov = ~product
)

summary(m_es_customs_base)
```

OLS estimation, Dep. Var.: ln_customs_value

Observations: 746,784

Fixed-effects: product: 9,668, date: 108

Standard-errors: Clustered (product)

	Estimate	Std. Error	t value	Pr(> t)
event_time::-60:highgap	-0.260643	0.139734	-1.865281	6.2172e-02 .
event_time::-59:highgap	-0.123196	0.140356	-0.877735	3.8011e-01
event_time::-58:highgap	-0.058897	0.138079	-0.426549	6.6972e-01
event_time::-57:highgap	-0.124774	0.140071	-0.890789	3.7306e-01
event_time::-56:highgap	-0.177143	0.138544	-1.278602	2.0107e-01
event_time::-55:highgap	0.126425	0.139528	0.906091	3.6491e-01
event_time::-54:highgap	0.240785	0.136085	1.769368	7.6864e-02 .
event_time::-53:highgap	0.325759	0.135311	2.407483	1.6082e-02 *
event_time::-52:highgap	0.444249	0.134992	3.290934	1.0021e-03 **
event_time::-51:highgap	0.536510	0.133629	4.014916	5.9916e-05 ***
event_time::-50:highgap	0.099337	0.133107	0.746296	4.5551e-01
event_time::-49:highgap	-0.041439	0.133308	-0.310850	7.5592e-01
event_time::-48:highgap	-0.084861	0.130140	-0.652071	5.1437e-01
event_time::-47:highgap	-0.020458	0.130802	-0.156404	8.7572e-01
event_time::-46:highgap	0.038305	0.129697	0.295339	7.6774e-01
event_time::-45:highgap	-0.081460	0.129502	-0.629021	5.2935e-01
event_time::-44:highgap	-0.165928	0.129656	-1.279757	2.0066e-01
event_time::-43:highgap	0.139217	0.130718	1.065014	2.8690e-01
event_time::-42:highgap	0.012670	0.131332	0.096470	9.2315e-01
event_time::-41:highgap	0.207443	0.130710	1.587048	1.1253e-01

event_time::-40:highgap	0.186442	0.130915	1.424139	1.5444e-01	
event_time::-39:highgap	0.131286	0.127490	1.029773	3.0314e-01	
event_time::-38:highgap	0.061469	0.125610	0.489367	6.2459e-01	
event_time::-37:highgap	-0.070922	0.129048	-0.549577	5.8262e-01	
event_time::-36:highgap	-0.139110	0.124034	-1.121546	2.6208e-01	
event_time::-35:highgap	0.044823	0.123605	0.362628	7.1689e-01	
event_time::-34:highgap	0.154222	0.124386	1.239862	2.1506e-01	
event_time::-33:highgap	-0.058631	0.124894	-0.469446	6.3876e-01	
event_time::-32:highgap	-0.012457	0.123499	-0.100870	9.1966e-01	
event_time::-31:highgap	0.158979	0.124196	1.280063	2.0055e-01	
event_time::-30:highgap	0.398777	0.124482	3.203483	1.3622e-03	**
event_time::-29:highgap	0.172639	0.125485	1.375775	1.6892e-01	
event_time::-28:highgap	0.299014	0.125322	2.385971	1.7053e-02	*
event_time::-27:highgap	0.396295	0.124888	3.173204	1.5124e-03	**
event_time::-26:highgap	0.160999	0.121270	1.327605	1.8434e-01	
event_time::-25:highgap	0.080103	0.121225	0.660785	5.0877e-01	
event_time::-24:highgap	0.161454	0.118702	1.360161	1.7381e-01	
event_time::-23:highgap	-0.025689	0.118799	-0.216239	8.2881e-01	
event_time::-22:highgap	0.044969	0.121289	0.370764	7.1082e-01	
event_time::-21:highgap	0.078650	0.120735	0.651430	5.1478e-01	
event_time::-20:highgap	0.120054	0.121237	0.990238	3.2208e-01	
event_time::-19:highgap	0.102365	0.119391	0.857397	3.9125e-01	
event_time::-18:highgap	0.179231	0.120143	1.491811	1.3578e-01	
event_time::-17:highgap	0.418571	0.119432	3.504665	4.5925e-04	***
event_time::-16:highgap	0.381650	0.120291	3.172713	1.5149e-03	**
event_time::-15:highgap	0.311196	0.118332	2.629859	8.5556e-03	**
event_time::-14:highgap	0.234519	0.114743	2.043856	4.0995e-02	*
event_time::-13:highgap	-0.010938	0.115529	-0.094674	9.2458e-01	
event_time::-12:highgap	0.042027	0.114847	0.365941	7.1442e-01	
event_time::-11:highgap	0.045401	0.113552	0.399821	6.8930e-01	
event_time::-10:highgap	0.099426	0.112815	0.881319	3.7817e-01	
event_time::-9:highgap	0.078617	0.111564	0.704682	4.8103e-01	
event_time::-8:highgap	0.023398	0.114296	0.204718	8.3780e-01	
event_time::-7:highgap	0.148869	0.112545	1.322743	1.8595e-01	
event_time::-6:highgap	0.180524	0.112529	1.604238	1.0869e-01	
event_time::-5:highgap	0.066217	0.112599	0.588082	5.5649e-01	
event_time::-4:highgap	0.359111	0.110182	3.259253	1.1209e-03	**
event_time::-3:highgap	0.264176	0.106219	2.487101	1.2896e-02	*
event_time::-2:highgap	0.307547	0.103258	2.978445	2.9043e-03	**
event_time:::0:highgap	0.141676	0.105040	1.348773	1.7744e-01	
event_time:::1:highgap	0.090050	0.105632	0.852483	3.9397e-01	
event_time:::2:highgap	0.025482	0.108507	0.234846	8.1433e-01	
event_time:::3:highgap	0.174958	0.109389	1.599409	1.0976e-01	

event_time::4:highgap	0.114835	0.112089	1.024499	3.0563e-01	
event_time::5:highgap	0.337267	0.112672	2.993353	2.7663e-03	**
event_time::6:highgap	0.417224	0.112227	3.717694	2.0219e-04	***
event_time::7:highgap	0.401166	0.112780	3.557056	3.7683e-04	***
event_time::8:highgap	0.499495	0.114296	4.370202	1.2543e-05	***
event_time::9:highgap	0.340040	0.113057	3.007684	2.6392e-03	**
event_time::10:highgap	0.324404	0.113262	2.864186	4.1898e-03	**
event_time::11:highgap	0.079404	0.110859	0.716258	4.7385e-01	
event_time::12:highgap	0.165008	0.111916	1.474386	1.4041e-01	
event_time::13:highgap	0.313396	0.111805	2.803044	5.0724e-03	**
event_time::14:highgap	0.162184	0.112628	1.440000	1.4990e-01	
event_time::15:highgap	0.140806	0.113778	1.237553	2.1591e-01	
event_time::16:highgap	0.221654	0.114293	1.939358	5.2487e-02	.
event_time::17:highgap	0.277576	0.116207	2.388634	1.6930e-02	*
event_time::18:highgap	0.228360	0.116902	1.953427	5.0798e-02	.
event_time::19:highgap	0.369947	0.115824	3.194044	1.4074e-03	**
event_time::20:highgap	0.317780	0.117856	2.696335	7.0229e-03	**
event_time::21:highgap	0.375317	0.117680	3.189311	1.4307e-03	**
event_time::22:highgap	0.288113	0.115206	2.500857	1.2406e-02	*
event_time::23:highgap	0.173061	0.116490	1.485636	1.3741e-01	
event_time::24:highgap	0.211534	0.114285	1.850933	6.4210e-02	.
event_time::25:highgap	0.101777	0.115728	0.879448	3.7918e-01	
event_time::26:highgap	0.104067	0.116790	0.891062	3.7292e-01	
event_time::27:highgap	0.173314	0.116477	1.487969	1.3679e-01	
event_time::28:highgap	0.139274	0.116301	1.197528	2.3113e-01	
event_time::29:highgap	0.198325	0.116286	1.705498	8.8134e-02	.
event_time::30:highgap	0.173340	0.117931	1.469843	1.4164e-01	
event_time::31:highgap	0.388423	0.118097	3.289021	1.0090e-03	**
event_time::32:highgap	0.379457	0.119557	3.173873	1.5089e-03	**
event_time::33:highgap	0.313218	0.119424	2.622753	8.7359e-03	**
event_time::34:highgap	0.299985	0.118103	2.540023	1.1100e-02	*
event_time::35:highgap	0.140744	0.118739	1.185323	2.3592e-01	
event_time::36:highgap	0.054082	0.117591	0.459919	6.4558e-01	
event_time::37:highgap	0.277648	0.118206	2.348856	1.8851e-02	*
event_time::38:highgap	0.070828	0.118335	0.598537	5.4950e-01	
event_time::39:highgap	0.137270	0.119145	1.152119	2.4930e-01	
event_time::40:highgap	0.109222	0.118091	0.924895	3.5504e-01	
event_time::41:highgap	0.306799	0.118785	2.582816	9.8143e-03	**
event_time::42:highgap	0.327422	0.120892	2.708395	6.7728e-03	**
event_time::43:highgap	0.320542	0.121065	2.647695	8.1174e-03	**
event_time::44:highgap	0.324999	0.121234	2.680751	7.3582e-03	**
event_time::45:highgap	0.196670	0.122101	1.610713	1.0728e-01	
event_time::46:highgap	0.198230	0.120661	1.642862	1.0044e-01	

```

event_time::47:highgap    0.163956    0.121891    1.345103    1.7862e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 3.49458      Adj. R2: 0.64575
                Within R2: 4.786e-4

```

```

out_customs_base <- file.path(figures_dir, "event_study_ln_customs_value.png")

plot_event_study_year(
  model_obj = m_es_customs_base,
  output_png = out_customs_base,
  y_label = "Effect on ln(1 + customs value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```

pdf
2

```
cat("Saved", out_customs_base, "\n")
```

Saved ../../results/figures/event_study/event_study_ln_customs_value.png

```

out_customs_base_guides <- file.path(seasonality_dir, "event_study_ln_customs_value_12mo_guides.png")
plot_event_study_year_guides(
  model_obj = m_es_customs_base,
  output_png = out_customs_base_guides,
  y_label = "Effect on ln(1 + customs value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```

pdf
2

```
cat("Saved", out_customs_base_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_ln_customs_value_12mo_guides.png


```

es_qty <- df_qty %>%
  filter(!is.na(event_time)) %>%
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)

cat("Rows in quantity event-study sample:", nrow(es_qty), "\n")

```

Rows in quantity event-study sample: 746784

```

cat("Quantity event-time range:", min(es_qty$event_time), "to", max(es_qty$event_time), "\n")

```

Quantity event-time range: -60 to 47

```

m_es_qty_base <- feols(
  ln_first_unit_qty ~ i(event_time, highgap, ref = -1) | product + date,
  data = es_qty,
  vcov = ~product
)

summary(m_es_qty_base)

```

OLS estimation, Dep. Var.: ln_first_unit_qty

Observations: 746,784

Fixed-effects: product: 9,668, date: 108

Standard-errors: Clustered (product)

	Estimate	Std. Error	t value	Pr(> t)
event_time::-60:highgap	0.033886	0.110598	0.306386	7.5932e-01
event_time::-59:highgap	0.207368	0.110625	1.874511	6.0890e-02 .
event_time::-58:highgap	0.280567	0.108837	2.577860	9.9561e-03 **
event_time::-57:highgap	0.231067	0.109390	2.112332	3.4684e-02 *
event_time::-56:highgap	0.164221	0.108818	1.509136	1.3130e-01
event_time::-55:highgap	0.368299	0.108898	3.382058	7.2230e-04 ***
event_time::-54:highgap	0.394248	0.107109	3.680810	2.3376e-04 ***
event_time::-53:highgap	0.393041	0.107592	3.653074	2.6049e-04 ***
event_time::-52:highgap	0.511189	0.105906	4.826813	1.4086e-06 ***
event_time::-51:highgap	0.518395	0.106254	4.878813	1.0843e-06 ***
event_time::-50:highgap	0.327838	0.104856	3.126541	1.7740e-03 **
event_time::-49:highgap	0.154245	0.104277	1.479180	1.3912e-01
event_time::-48:highgap	0.206099	0.102827	2.004320	4.5064e-02 *
event_time::-47:highgap	0.304422	0.103748	2.934245	3.3515e-03 **
event_time::-46:highgap	0.326255	0.103538	3.151081	1.6316e-03 **

event_time::-45:highgap	0.278495	0.103307	2.695803	7.0341e-03	**
event_time::-44:highgap	0.127186	0.104387	1.218401	2.2310e-01	
event_time::-43:highgap	0.353549	0.104364	3.387659	7.0773e-04	***
event_time::-42:highgap	0.213479	0.103972	2.053237	4.0076e-02	*
event_time::-41:highgap	0.365660	0.104790	3.489446	4.8618e-04	***
event_time::-40:highgap	0.397584	0.102946	3.862060	1.1317e-04	***
event_time::-39:highgap	0.233812	0.102146	2.289003	2.2101e-02	*
event_time::-38:highgap	0.254071	0.099719	2.547853	1.0854e-02	*
event_time::-37:highgap	0.127416	0.102197	1.246771	2.1251e-01	
event_time::-36:highgap	0.082241	0.098634	0.833801	4.0441e-01	
event_time::-35:highgap	0.253615	0.098848	2.565719	1.0311e-02	*
event_time::-34:highgap	0.314354	0.097435	3.226276	1.2583e-03	**
event_time::-33:highgap	0.158025	0.099344	1.590678	1.1171e-01	
event_time::-32:highgap	0.140817	0.098806	1.425180	1.5414e-01	
event_time::-31:highgap	0.281163	0.098947	2.841555	4.4988e-03	**
event_time::-30:highgap	0.459257	0.098286	4.672643	3.0137e-06	***
event_time::-29:highgap	0.272300	0.098816	2.755643	5.8687e-03	**
event_time::-28:highgap	0.369626	0.099643	3.709507	2.0883e-04	***
event_time::-27:highgap	0.415868	0.098823	4.208188	2.5975e-05	***
event_time::-26:highgap	0.247629	0.096590	2.563715	1.0371e-02	*
event_time::-25:highgap	0.213769	0.095356	2.241809	2.4996e-02	*
event_time::-24:highgap	0.269946	0.093747	2.879506	3.9917e-03	**
event_time::-23:highgap	0.142372	0.093797	1.517872	1.2908e-01	
event_time::-22:highgap	0.157817	0.095113	1.659270	9.7094e-02	.
event_time::-21:highgap	0.169327	0.095795	1.767583	7.7162e-02	.
event_time::-20:highgap	0.112365	0.096414	1.165438	2.4387e-01	
event_time::-19:highgap	0.212765	0.094711	2.246463	2.4697e-02	*
event_time::-18:highgap	0.213183	0.096139	2.217437	2.6616e-02	*
event_time::-17:highgap	0.366284	0.093983	3.897356	9.7905e-05	***
event_time::-16:highgap	0.373189	0.094575	3.945957	8.0044e-05	***
event_time::-15:highgap	0.334547	0.094457	3.541801	3.9928e-04	***
event_time::-14:highgap	0.245634	0.090457	2.715480	6.6297e-03	**
event_time::-13:highgap	0.086992	0.090559	0.960609	3.3677e-01	
event_time::-12:highgap	0.097935	0.089692	1.091903	2.7490e-01	
event_time::-11:highgap	0.110786	0.087984	1.259153	2.0801e-01	
event_time::-10:highgap	0.111655	0.088563	1.260749	2.0743e-01	
event_time::-9:highgap	0.096702	0.087584	1.104104	2.6958e-01	
event_time::-8:highgap	0.082469	0.088683	0.929934	3.5243e-01	
event_time::-7:highgap	0.149769	0.087511	1.711420	8.7036e-02	.
event_time::-6:highgap	0.191871	0.087620	2.189814	2.8562e-02	*
event_time::-5:highgap	0.108424	0.086549	1.252739	2.1033e-01	
event_time::-4:highgap	0.350828	0.084601	4.146871	3.3993e-05	***
event_time::-3:highgap	0.231201	0.081917	2.822394	4.7764e-03	**

event_time::-2:highgap	0.261361	0.080248	3.256903	1.1302e-03	**
event_time::0:highgap	0.171095	0.081602	2.096695	3.6047e-02	*
event_time::1:highgap	0.112855	0.082629	1.365802	1.7203e-01	
event_time::2:highgap	0.120816	0.084668	1.426945	1.5363e-01	
event_time::3:highgap	0.197079	0.085248	2.311838	2.0808e-02	*
event_time::4:highgap	0.127751	0.087946	1.452601	1.4637e-01	
event_time::5:highgap	0.285650	0.087529	3.263492	1.1043e-03	**
event_time::6:highgap	0.296602	0.088350	3.357111	7.9065e-04	***
event_time::7:highgap	0.329426	0.089436	3.683387	2.3141e-04	***
event_time::8:highgap	0.349563	0.089630	3.900065	9.6818e-05	***
event_time::9:highgap	0.307714	0.089160	3.451257	5.6037e-04	***
event_time::10:highgap	0.300215	0.089154	3.367387	7.6180e-04	***
event_time::11:highgap	0.013714	0.087167	0.157334	8.7499e-01	
event_time::12:highgap	0.115589	0.088508	1.305981	1.9159e-01	
event_time::13:highgap	0.240734	0.088089	2.732852	6.2903e-03	**
event_time::14:highgap	0.158830	0.089448	1.775673	7.5818e-02	.
event_time::15:highgap	0.119954	0.090545	1.324793	1.8527e-01	
event_time::16:highgap	0.186884	0.091152	2.050242	4.0368e-02	*
event_time::17:highgap	0.221974	0.092968	2.387639	1.6976e-02	*
event_time::18:highgap	0.148807	0.093266	1.595513	1.1063e-01	
event_time::19:highgap	0.298331	0.091866	3.247474	1.1683e-03	**
event_time::20:highgap	0.233497	0.093949	2.485372	1.2958e-02	*
event_time::21:highgap	0.320116	0.094395	3.391253	6.9853e-04	***
event_time::22:highgap	0.214582	0.091994	2.332569	1.9691e-02	*
event_time::23:highgap	0.113859	0.093391	1.219164	2.2281e-01	
event_time::24:highgap	0.182784	0.091474	1.998210	4.5722e-02	*
event_time::25:highgap	0.138151	0.092160	1.499033	1.3390e-01	
event_time::26:highgap	0.093854	0.093127	1.007810	3.1357e-01	
event_time::27:highgap	0.134083	0.093184	1.438903	1.5021e-01	
event_time::28:highgap	0.074600	0.093630	0.796758	4.2561e-01	
event_time::29:highgap	0.142458	0.094813	1.502520	1.3300e-01	
event_time::30:highgap	0.147530	0.094755	1.556958	1.1951e-01	
event_time::31:highgap	0.255901	0.095030	2.692853	7.0966e-03	**
event_time::32:highgap	0.188491	0.096381	1.955696	5.0530e-02	.
event_time::33:highgap	0.169130	0.096008	1.761625	7.8164e-02	.
event_time::34:highgap	0.150875	0.094336	1.599335	1.0978e-01	
event_time::35:highgap	0.004432	0.095708	0.046312	9.6306e-01	
event_time::36:highgap	-0.003581	0.094846	-0.037755	9.6988e-01	
event_time::37:highgap	0.156822	0.095214	1.647057	9.9579e-02	.
event_time::38:highgap	0.021222	0.095175	0.222981	8.2355e-01	
event_time::39:highgap	0.069062	0.095708	0.721585	4.7057e-01	
event_time::40:highgap	-0.042989	0.094866	-0.453155	6.5045e-01	
event_time::41:highgap	0.131619	0.097015	1.356691	1.7491e-01	

```

event_time::42:highgap    0.140794    0.098137    1.434670 1.5141e-01
event_time::43:highgap    0.157260    0.098604    1.594868 1.1077e-01
event_time::44:highgap    0.154671    0.098717    1.566809 1.1719e-01
event_time::45:highgap    0.083109    0.098646    0.842499 3.9953e-01
event_time::46:highgap    0.036569    0.097201    0.376225 7.0676e-01
event_time::47:highgap    0.037741    0.098883    0.381667 7.0272e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.80244      Adj. R2: 0.730807
                Within R2: 3.969e-4

```

```

out_qty_base <- file.path(figures_dir, "event_study_ln_first_unit_qty.png")

plot_event_study_year(
  model_obj = m_es_qty_base,
  output_png = out_qty_base,
  y_label = "Effect on ln(1 + first unit quantity)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```

pdf
2

```
cat("Saved", out_qty_base, "\n")
```

Saved ../../results/figures/event_study/event_study_ln_first_unit_qty.png

```

out_qty_base_guides <- file.path(seasonality_dir, "event_study_ln_first_unit_qty_12mo_guides.png")
plot_event_study_year_guides(
  model_obj = m_es_qty_base,
  output_png = out_qty_base_guides,
  y_label = "Effect on ln(1 + first unit quantity)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```

pdf
2

```
cat("Saved", out_qty_base_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_ln_first_unit_qty_12mo_guides.png

```
es_uv <- df_uv %>%  
  filter(!is.na(event_time)) %>%  
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)  
  
cat("Rows in unit-value event-study sample:", nrow(es_uv), "\n")
```

Rows in unit-value event-study sample: 418690

```
cat("Unit-value event-time range:", min(es_uv$event_time), "to", max(es_uv$event_time), "\n")
```

Unit-value event-time range: -60 to 47

```
m_es_uv_base <- feols(  
  ln_unit_value ~ i(event_time, highgap, ref = -1) | product + date,  
  data = es_uv,  
  vcov = ~product  
)  
  
summary(m_es_uv_base)
```

OLS estimation, Dep. Var.: ln_unit_value

Observations: 418,184

Fixed-effects: product: 8,017, date: 108

Standard-errors: Clustered (product)

	Estimate	Std. Error	t value	Pr(> t)	
event_time::-60:highgap	-0.093952	0.042371	-2.217361	0.0266264	*
event_time::-59:highgap	-0.081305	0.044858	-1.812486	0.0699485	.
event_time::-58:highgap	0.016621	0.045867	0.362373	0.7170829	
event_time::-57:highgap	-0.093652	0.043172	-2.169265	0.0300918	*
event_time::-56:highgap	-0.056066	0.044004	-1.274130	0.2026543	
event_time::-55:highgap	-0.084896	0.044159	-1.922479	0.0545809	.
event_time::-54:highgap	-0.095098	0.043257	-2.198423	0.0279474	*
event_time::-53:highgap	-0.060484	0.043695	-1.384241	0.1663232	
event_time::-52:highgap	-0.041604	0.043745	-0.951072	0.3415966	
event_time::-51:highgap	-0.070016	0.042644	-1.641861	0.1006581	

event_time::-50:highgap	-0.013404	0.043436	-0.308601	0.7576333	
event_time::-49:highgap	-0.055203	0.043040	-1.282600	0.1996693	
event_time::-48:highgap	-0.038946	0.042365	-0.919281	0.3579761	
event_time::-47:highgap	-0.089077	0.043920	-2.028194	0.0425735	*
event_time::-46:highgap	-0.035993	0.045520	-0.790704	0.4291399	
event_time::-45:highgap	-0.043584	0.042862	-1.016863	0.3092492	
event_time::-44:highgap	-0.063861	0.042411	-1.505748	0.1321714	
event_time::-43:highgap	-0.082445	0.041233	-1.999490	0.0455891	*
event_time::-42:highgap	-0.009653	0.042874	-0.225150	0.8218683	
event_time::-41:highgap	-0.041946	0.040904	-1.025474	0.3051703	
event_time::-40:highgap	-0.080541	0.042143	-1.911134	0.0560230	.
event_time::-39:highgap	-0.000557	0.041722	-0.013349	0.9893495	
event_time::-38:highgap	-0.089807	0.042022	-2.137151	0.0326159	*
event_time::-37:highgap	-0.033834	0.043984	-0.769252	0.4417661	
event_time::-36:highgap	-0.059690	0.041881	-1.425230	0.1541298	
event_time::-35:highgap	-0.014315	0.040045	-0.357475	0.7207459	
event_time::-34:highgap	-0.004798	0.041760	-0.114900	0.9085274	
event_time::-33:highgap	-0.033929	0.042020	-0.807441	0.4194366	
event_time::-32:highgap	0.001528	0.039713	0.038487	0.9693001	
event_time::-31:highgap	-0.022745	0.040318	-0.564130	0.5726812	
event_time::-30:highgap	0.000465	0.040497	0.011471	0.9908483	
event_time::-29:highgap	-0.043002	0.039175	-1.097708	0.2723650	
event_time::-28:highgap	-0.000101	0.041131	-0.002458	0.9980388	
event_time::-27:highgap	0.011485	0.039989	0.287208	0.7739602	
event_time::-26:highgap	-0.037536	0.040358	-0.930071	0.3523626	
event_time::-25:highgap	-0.021967	0.039778	-0.552246	0.5807950	
event_time::-24:highgap	-0.068668	0.039885	-1.721631	0.0851750	.
event_time::-23:highgap	-0.065354	0.041284	-1.583046	0.1134504	
event_time::-22:highgap	-0.037760	0.038825	-0.972577	0.3307930	
event_time::-21:highgap	-0.001219	0.038932	-0.031305	0.9750274	
event_time::-20:highgap	-0.005999	0.039421	-0.152184	0.8790459	
event_time::-19:highgap	-0.032179	0.039795	-0.808610	0.4187634	
event_time::-18:highgap	0.000650	0.037769	0.017215	0.9862653	
event_time::-17:highgap	0.004694	0.038624	0.121543	0.9032640	
event_time::-16:highgap	-0.004196	0.038783	-0.108189	0.9138487	
event_time::-15:highgap	-0.064192	0.039143	-1.639937	0.1010575	
event_time::-14:highgap	-0.040487	0.038888	-1.041102	0.2978594	
event_time::-13:highgap	-0.048805	0.037000	-1.319064	0.1871853	
event_time::-12:highgap	-0.011084	0.036361	-0.304829	0.7605042	
event_time::-11:highgap	0.031859	0.037154	0.857468	0.3912118	
event_time::-10:highgap	0.015124	0.037163	0.406972	0.6840394	
event_time::-9:highgap	-0.009559	0.037568	-0.254445	0.7991587	
event_time::-8:highgap	0.012995	0.035960	0.361372	0.7178307	

event_time::-7:highgap	0.063714	0.036543	1.743536	0.0812784	.
event_time::-6:highgap	-0.012306	0.035508	-0.346585	0.7289123	
event_time::-5:highgap	0.048202	0.035388	1.362125	0.1731967	
event_time::-4:highgap	-0.003114	0.036630	-0.085003	0.9322611	
event_time::-3:highgap	-0.000975	0.034588	-0.028203	0.9775008	
event_time::-2:highgap	0.036517	0.034985	1.043798	0.2966101	
event_time::0:highgap	-0.053038	0.036408	-1.456762	0.1452214	
event_time::1:highgap	-0.043464	0.036418	-1.193465	0.2327226	
event_time::2:highgap	-0.028725	0.036165	-0.794259	0.4270685	
event_time::3:highgap	-0.056833	0.037389	-1.520039	0.1285406	
event_time::4:highgap	-0.041896	0.036163	-1.158536	0.2466800	
event_time::5:highgap	-0.021334	0.036277	-0.588092	0.5564874	
event_time::6:highgap	0.009324	0.036197	0.257580	0.7967375	
event_time::7:highgap	-0.053268	0.036005	-1.479447	0.1390603	
event_time::8:highgap	0.032749	0.037015	0.884758	0.3763136	
event_time::9:highgap	-0.024676	0.036576	-0.674650	0.4999175	
event_time::10:highgap	-0.022179	0.038383	-0.577819	0.5634023	
event_time::11:highgap	-0.034338	0.036622	-0.937627	0.3484645	
event_time::12:highgap	-0.017906	0.036124	-0.495687	0.6201286	
event_time::13:highgap	-0.023173	0.037794	-0.613151	0.5397938	
event_time::14:highgap	-0.011800	0.036406	-0.324111	0.7458622	
event_time::15:highgap	-0.039120	0.035997	-1.086759	0.2771759	
event_time::16:highgap	-0.067271	0.037224	-1.807205	0.0707677	.
event_time::17:highgap	-0.053511	0.037630	-1.422034	0.1550554	
event_time::18:highgap	-0.017318	0.037416	-0.462855	0.6434808	
event_time::19:highgap	-0.051291	0.038367	-1.336842	0.1813122	
event_time::20:highgap	-0.019190	0.038000	-0.504988	0.6135810	
event_time::21:highgap	-0.067493	0.037999	-1.776178	0.0757416	.
event_time::22:highgap	-0.036880	0.037461	-0.984494	0.3249026	
event_time::23:highgap	-0.038552	0.038744	-0.995056	0.3197389	
event_time::24:highgap	-0.055615	0.037538	-1.481563	0.1384959	
event_time::25:highgap	-0.023252	0.038588	-0.602563	0.5468164	
event_time::26:highgap	-0.040225	0.037723	-1.066342	0.2863011	
event_time::27:highgap	-0.063857	0.037754	-1.691400	0.0907993	.
event_time::28:highgap	-0.073730	0.038019	-1.939280	0.0525023	.
event_time::29:highgap	-0.069068	0.038351	-1.800943	0.0717494	.
event_time::30:highgap	-0.088982	0.036924	-2.409844	0.0159817	*
event_time::31:highgap	-0.083024	0.037311	-2.225164	0.0260979	*
event_time::32:highgap	-0.025422	0.036992	-0.687241	0.4919506	
event_time::33:highgap	-0.037493	0.038080	-0.984596	0.3248524	
event_time::34:highgap	-0.058859	0.038461	-1.530356	0.1259680	
event_time::35:highgap	-0.046145	0.038150	-1.209589	0.2264724	
event_time::36:highgap	-0.059575	0.037981	-1.568559	0.1167903	

```

event_time::37:highgap -0.087539 0.038280 -2.286807 0.0222330 *
event_time::38:highgap -0.112551 0.037816 -2.976281 0.0029264 **
event_time::39:highgap -0.113572 0.037838 -3.001568 0.0026942 **
event_time::40:highgap -0.055244 0.037495 -1.473352 0.1406954
event_time::41:highgap -0.087947 0.037948 -2.317593 0.0204965 *
event_time::42:highgap -0.092226 0.037582 -2.453960 0.0141504 *
event_time::43:highgap -0.104425 0.038285 -2.727601 0.0063936 **
event_time::44:highgap -0.086667 0.039318 -2.204267 0.0275339 *
event_time::45:highgap -0.110529 0.037977 -2.910410 0.0036195 **
event_time::46:highgap -0.068573 0.038429 -1.784408 0.0743952 .
event_time::47:highgap -0.060120 0.038020 -1.581252 0.1138599

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.821171 Adj. R2: 0.868466

Within R2: 4.842e-4

```

out_uv_base <- file.path(figures_dir, "event_study_ln_unit_value.png")

```

```

plot_event_study_year(
  model_obj = m_es_uv_base,
  output_png = out_uv_base,
  y_label = "Effect on ln(unit value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```

pdf

2

```

cat("Saved", out_uv_base, "\n")

```

Saved ../../results/figures/event_study/event_study_ln_unit_value.png

```

out_uv_base_guides <- file.path(seasonality_dir, "event_study_ln_unit_value_12mo_guides.png")
plot_event_study_year_guides(
  model_obj = m_es_uv_base,
  output_png = out_uv_base_guides,
  y_label = "Effect on ln(unit value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST
)

```


pdf
2

```
cat("Saved", out_uv_base_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_ln_unit_value_12mo_guides.png

Continuous-Treatment Event Study (Gap Intensity)

```
d_cont <- df %>%  
  filter(!is.na(gap_pre))  
  
d_unit_consistent <- d_cont %>%  
  filter(unit_consistent == 1)  
  
cat("Rows in continuous-treatment base sample:", nrow(d_cont), "\n")
```

Rows in continuous-treatment base sample: 766524

```
cat("Unique products in continuous-treatment base sample:", n_distinct(d_cont$product), "\n")
```

Unique products in continuous-treatment base sample: 8621

```
cat("Rows in continuous-treatment unit-consistent sample:", nrow(d_unit_consistent), "\n")
```

Rows in continuous-treatment unit-consistent sample: 766524

```
cat("Unique products in continuous-treatment unit-consistent sample:", n_distinct(d_unit_consistent$product), "\n")
```

Unique products in continuous-treatment unit-consistent sample: 8621

```
es_customs_cont <- d_cont %>%  
  filter(!is.na(ln_customs_value), !is.na(event_time)) %>%  
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)  
  
cat("Rows in customs continuous event-study sample:", nrow(es_customs_cont), "\n")
```

Rows in customs continuous event-study sample: 702180

```
cat("Customs continuous event-time range:", min(es_customs_cont$event_time), "to", max(es_customs_cont$event_time))
```

Customs continuous event-time range: -60 to 47

```
m_es_customs_cont <- feols(
  ln_customs_value ~ i(event_time, gap_pre, ref = -1) | product + date,
  data = es_customs_cont,
  vcov = ~product
)

pt_customs_cont <- wald(m_es_customs_cont, keep = "^event_time::-[0-9]+:gap_pre$")
```

Wald test, H0: joint nullity of event_time::-60:gap_pre, event_time::-59:gap_pre, event_time
stat = 2.35253, p-value = 2.235e-8, on 59 and 693,375 DoF, VCOV: Clustered (product).

```
cat("Customs pre-trend joint test (k < 0):\n")
```

Customs pre-trend joint test ($k < 0$):

```
print(pt_customs_cont)
```

```
$stat
[1] 2.352532
```

```
$p
[1] 2.234659e-08
```

```
$df1
[1] 59
```

```
$df2
[1] 693375
```

```
$vcov
[1] "Clustered (product)"
```

```

out_customs_cont <- file.path(figures_dir, "event_study_continuous_ln_customs_value.png")
plot_event_study_year(
  model_obj = m_es_customs_cont,
  output_png = out_customs_cont,
  y_label = "Effect per +1 NTR gap on ln(1 + customs value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_customs_cont, "\n")
```

Saved ../../results/figures/event_study/event_study_continuous_ln_customs_value.png

```

out_customs_cont_guides <- file.path(seasonality_dir, "event_study_continuous_ln_customs_val
plot_event_study_year_guides(
  model_obj = m_es_customs_cont,
  output_png = out_customs_cont_guides,
  y_label = "Effect per +1 NTR gap on ln(1 + customs value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_customs_cont_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_continuous_ln_customs_value_12mo_guides.png

```

es_qty_cont <- d_unit_consistent %>%
  filter(!is.na(ln_first_unit_qty), !is.na(event_time)) %>%
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)

cat("Rows in quantity continuous event-study sample:", nrow(es_qty_cont), "\n")

```

Rows in quantity continuous event-study sample: 702180

```
cat("Quantity continuous event-time range:", min(es_qty_cont$event_time), "to", max(es_qty_c
```

Quantity continuous event-time range: -60 to 47

```
m_es_qty_cont <- feols(  
  ln_first_unit_qty ~ i(event_time, gap_pre, ref = -1) | product + date,  
  data = es_qty_cont,  
  vcov = ~product  
)  
  
pt_qty_cont <- wald(m_es_qty_cont, keep = "^event_time::-[0-9]+:gap_pre$")
```

Wald test, H0: joint nullity of event_time::-60:gap_pre, event_time::-59:gap_pre, event_time
stat = 2.16948, p-value = 5.197e-7, on 59 and 693,375 DoF, VCOV: Clustered (product).

```
cat("Quantity pre-trend joint test (k < 0):\n")
```

Quantity pre-trend joint test (k < 0):

```
print(pt_qty_cont)
```

\$stat

[1] 2.169483

\$p

[1] 5.196943e-07

\$df1

[1] 59

\$df2

[1] 693375

\$vcov

[1] "Clustered (product)"

```

out_qty_cont <- file.path(figures_dir, "event_study_continuous_ln_first_unit_qty.png")
plot_event_study_year(
  model_obj = m_es_qty_cont,
  output_png = out_qty_cont,
  y_label = "Effect per +1 NTR gap on ln(1 + first unit quantity)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_qty_cont, "\n")
```

Saved ../../results/figures/event_study/event_study_continuous_ln_first_unit_qty.png

```

out_qty_cont_guides <- file.path(seasonality_dir, "event_study_continuous_ln_first_unit_qty_12mo_guides.png")
plot_event_study_year_guides(
  model_obj = m_es_qty_cont,
  output_png = out_qty_cont_guides,
  y_label = "Effect per +1 NTR gap on ln(1 + first unit quantity)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_qty_cont_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_continuous_ln_first_unit_qty_12mo_guides.png

```

es_uv_cont <- d_unit_consistent %>%
  filter(!is.na(ln_unit_value), !is.na(event_time)) %>%
  filter(event_time >= -WIN_PRE, event_time <= WIN_POST)

cat("Rows in unit-value continuous event-study sample:", nrow(es_uv_cont), "\n")

```

Rows in unit-value continuous event-study sample: 395259

```
cat("Unit-value continuous event-time range:", min(es_uv_cont$event_time), "to", max(es_uv_c
```

Unit-value continuous event-time range: -60 to 47

```
m_es_uv_cont <- feols(  
  ln_unit_value ~ i(event_time, gap_pre, ref = -1) | product + date,  
  data = es_uv_cont,  
  vcov = ~product  
)  
  
pt_uv_cont <- wald(m_es_uv_cont, keep = "^event_time::-[0-9]+:gap_pre$")
```

Wald test, H0: joint nullity of event_time::-60:gap_pre, event_time::-59:gap_pre, event_time
stat = 1.26733, p-value = 0.080807, on 59 and 387,482 DoF, VCOV: Clustered (product).

```
cat("Unit-value pre-trend joint test (k < 0):\n")
```

Unit-value pre-trend joint test (k < 0):

```
print(pt_uv_cont)
```

\$stat

[1] 1.26733

\$p

[1] 0.08080661

\$df1

[1] 59

\$df2

[1] 387482

\$vcov

[1] "Clustered (product)"

```

out_uv_cont <- file.path(figures_dir, "event_study_continuous_ln_unit_value.png")
plot_event_study_year(
  model_obj = m_es_uv_cont,
  output_png = out_uv_cont,
  y_label = "Effect per +1 NTR gap on ln(unit value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_uv_cont, "\n")
```

Saved ../../results/figures/event_study/event_study_continuous_ln_unit_value.png

```

out_uv_cont_guides <- file.path(seasonality_dir, "event_study_continuous_ln_unit_value_12mo_guides.png")
plot_event_study_year_guides(
  model_obj = m_es_uv_cont,
  output_png = out_uv_cont_guides,
  y_label = "Effect per +1 NTR gap on ln(unit value)",
  win_pre = WIN_PRE,
  win_post = WIN_POST,
  term_name = "gap_pre"
)

```

pdf
2

```
cat("Saved", out_uv_cont_guides, "\n")
```

Saved ../../results/figures/seasonality/event_study_continuous_ln_unit_value_12mo_guides.png

```

out_cont_es_models <- file.path(tables_dir, "event_study_continuous_gap_pre_models.txt")
capture.output(
  etable(m_es_customs_cont, m_es_qty_cont, m_es_uv_cont),
  file = out_cont_es_models
)
cat("Saved", out_cont_es_models, "\n")

```

Saved ../../results/tables/event_study/event_study_continuous_gap_pre_models.txt

```
cont_pretrend <- tibble(  
  outcome = c("ln_customs_value", "ln_first_unit_qty", "ln_unit_value"),  
  stat = c(pt_customs_cont$stat, pt_qty_cont$stat, pt_uv_cont$stat),  
  p_value = c(pt_customs_cont$p, pt_qty_cont$p, pt_uv_cont$p),  
  df1 = c(pt_customs_cont$df1, pt_qty_cont$df1, pt_uv_cont$df1),  
  df2 = c(pt_customs_cont$df2, pt_qty_cont$df2, pt_uv_cont$df2),  
  n_obs = c(nobs(m_es_customs_cont), nobs(m_es_qty_cont), nobs(m_es_uv_cont))  
)  
print(cont_pretrend)
```

A tibble: 3 x 6

	outcome <chr>	stat <dbl>	p_value <dbl>	df1 <int>	df2 <dbl>	n_obs <int>
1	ln_customs_value	2.35	0.0000000223	59	693375	702180
2	ln_first_unit_qty	2.17	0.000000520	59	693375	702180
3	ln_unit_value	1.27	0.0808	59	387482	394851

```
out_cont_es_pretrend <- file.path(tables_dir, "event_study_continuous_gap_pre_pretrend_tests")  
write_csv(cont_pretrend, out_cont_es_pretrend, na = "")  
cat("Saved", out_cont_es_pretrend, "\n")
```

Saved ../../results/tables/event_study/event_study_continuous_gap_pre_pretrend_tests.csv